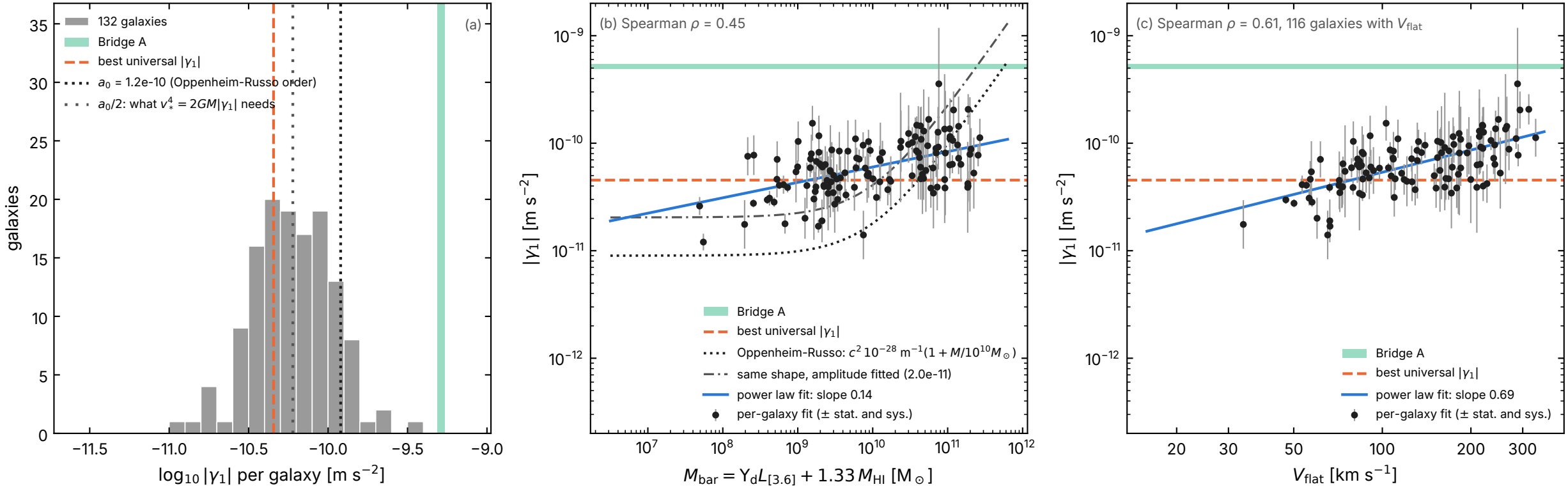
(a)
132 galaxies
Bridge A
best universal |γ1|
a0 = 1.2e-10 (Oppenheim-Russo order)
a0/2: what v*^4 = 2GM|γ1| needs
galaxies
log10 |γ1| per galaxy [m s−2]
(b) Spearman ρ = 0.45
Bridge A
best universal |γ1|
Oppenheim-Russo: c² 10⁻²⁸ m⁻¹(1 + M/10¹⁰M⊙)
same shape, amplitude fitted (2.0e-11)
power law fit: slope 0.14
per-galaxy fit (± stat. and sys.)
|γ1| [m s−2]
Mbar = YdL[3.6] + 1.33 MHI [M⊙]
(c) Spearman ρ = 0.61, 116 galaxies with Vflat
Bridge A
best universal |γ1|
power law fit: slope 0.69
per-galaxy fit (± stat. and sys.)
|γ1| [m s−2]
Vflat [km s−1]